

Module Overview

Essential Question: How can learning about natural disasters make us safer?

Quote: "Yes, a dark time passed over this land, but now there is something like light." — Dave Eggers

Big Idea Words (4): notable, tremor, spontaneous, hazard

Main Selections (workbook 4):

- Eruption! Volcanoes and the Science of Saving Lives · Elizabeth Rusch (photographs by Tom Uhlman) · Narrative Nonfiction (RI)
- Between the Glacier and the Sea: The Alaska Earthquake · from the Valdez Museum · Media: Video (RI)
- Quaking Earth, Racing Waves · Rachel Young · Informational Text (RI)
- Hurricanes: The Science Behind Killer Storms · Alvin and Virginia Silverstein and Laura Silverstein Nunn · Informational Text (RI)

Note. The SoT lists 31 core vocabulary items on Worksheet A (4 Big Idea + 8 Eruption! + 4 Between the Glacier and the Sea + 7 Quaking Earth, Racing Waves + 8 Hurricanes), in myBook appearance order. This G5 Teacher's Guide labels each selection's panel "Critical Vocabulary" — the string "Power Words" does not appear in the source. The teacher "Vocabulary Strategy" boxes (Eruption!: magma/wrought/stratosphere/muttered; Quaking Earth: plates/fault; Hurricanes: stages/sinks) and inline LANGUAGE craft notes (idioms; multiple-meaning words such as coral, drill, eye; domain terms such as eyewall) are NOT counted as core vocabulary. "Who Studies Natural Disasters?" is the Big-Idea Short Read / Letter (no Critical Vocabulary panel) and carries the 4 Big Idea Words; it is excluded from the per-selection workbook units. This module is ENTIRELY INFORMATIONAL — all four vocabulary-bearing main selections are informational (RI): a narrative-nonfiction volcano account, an informational video, and two informational science texts.

Answer Distribution (83 Multiple-Choice Items)

Verified counts across Worksheets A (31), B (32), and D (20). Worksheet C is open-response IB and not counted in MCQ balance.

A · 21 · B · 21 · C · 21 · D · 20

These four totals are the count of correct-answer LETTERS (A/B/C/D) across all 83 multiple-choice items — exactly 21 are "A", 21 are "B", 21 are "C", and 20 are "D". Within each worksheet: Worksheet A = 8 A / 8 B / 8 C / 7 D (31), Worksheet B = 8 / 8 / 8 / 8 (32), Worksheet D = 5 / 5 / 5 / 5 (20). Balance is achieved via choice-order rotation only; correct-answer meanings and option texts are preserved from the Source of Truth. All 31 SoT vocabulary words are covered on Worksheet A in myBook appearance order (Big Idea → Eruption! → Between the Glacier and the Sea → Quaking Earth, Racing Waves → Hurricanes).

Voca Trap Items (Worksheet A)

9 of 31 = 29%, all flagged below. Trap pattern: opposite-meaning / wrong-field baits drawn from the natural meaning field of each word — no hallucination.

- Q3 **spontaneous** — opposite bait ("planned weeks ahead of time," "follows a careful written schedule") vs. "happens naturally and isn't planned"
- Q9 **consequences** — wrong-direction bait ("the causes that come before events") vs. "the outcomes or effects of events"
- Q11 **alarming** — opposite bait ("makes you feel calm and safe") vs. "makes you worry that something bad may happen"
- Q13 **prior** — opposite bait ("coming after everything else") vs. "coming before, in time or order"
- Q15 **literally** — opposite bait ("only a make-believe story," "an exaggeration") vs. "it actually happened, and you aren't exaggerating"
- Q19 **destruction** — opposite bait ("the act of building something new") vs. "the act of destroying or ruining something"
- Q23 **modified** — opposite bait ("an exact copy with nothing different," "the very first version") vs. "a revised, or changed, version"
- Q25 **eventually** — opposite bait ("at this very second," "never happen at all") vs. "at some time, usually after a series of other events"

- Q28 **anchored** — opposite bait ("loose and free to roll away," "floating high in the air") vs. "firmly attached so it won't move easily"

Reading Trap Items (Worksheet B)

1 of 32 = 3%, flagged below. Trap pattern for Section B: central-idea bait (a true detail mistaken for the central idea) with length-balanced correct answers so a longest-option-wins guess fails. Distractors built from the SoT only — no hallucination. All four anchors are INFORMATIONAL (RI): Eruption! (Narrative Nonfiction), Between the Glacier and the Sea (Media: Video), Quaking Earth, Racing Waves (Informational Text), and Hurricanes (Informational Text). Codes used are RI.5.1 / .2 / .3 / .5 / .6 / .8. **RI.5.4 and RI.5.7 are NOT used** (and no RL codes appear, because this module has no literary main selection), and there are no "from the text" snippet boxes. The single Section B central-idea trap is on the Hurricanes selection.

- Q32 **Hurricanes central idea** — "a hurricane has an eye at its center," "some hurricanes are given names," "a hurricane scale goes up to Category 5" are true details, baited against the real central idea: understanding how hurricanes form and what damage they cause helps people prepare and stay safe

Because all four selections are informational, Section B is tested with central idea, key details, relationships among events/ideas, text structure (RI.5.5), author's purpose/point of view (RI.5.6), and reasons-and-evidence (RI.5.8) items. RI.5.8 fits the cause-and-effect, reasons-and-evidence structure of Eruption!, Quaking Earth, and Hurricanes especially well. Every Section B distractor is grounded in the Source of Truth.



How Every Item Is Grounded

Every item on this Answer Key traces to the Source-of-Truth JSON for G5 Module 3, built from the Teaching Pal OCR (4,357 lines, 34 spread-pages) and verified against the PDF intro spreads (every Critical Vocabulary panel was read from a high-resolution crop, and each panel's rounded bottom corner was confirmed so no words are cut off). No characters, words, events, statistics, or dates outside the Source of Truth were added.

- **Vocabulary (Worksheet A):** all 31 words come from the module's Big Idea Words (4: notable, tremor, spontaneous, hazard) and the four "Critical Vocabulary" panels — Eruption! 8, Between the Glacier and the Sea 4, Quaking Earth, Racing Waves 7, Hurricanes 8. Each meaning is taken from the surviving inline myNotes glosses printed in the body text (this M3 OCR has NO back-of-book Glossary; the file ends at the Performance Task), checked against the visually-read panels. The teacher Vocabulary Strategy boxes and the craft notes (idioms, multiple-meaning words, domain terms) are deliberately NOT counted as core vocabulary.
- **Terminology:** this G5 Teacher's Guide labels each panel "Critical Vocabulary," not "Power Words" — the workbook uses that exact label, matching the source.
- **Reading (Worksheet B):** the 32 items use only RI.5 codes, because all four anchors are informational (Eruption!, Between the Glacier and the Sea, Quaking Earth Racing Waves, Hurricanes). Each correct answer is grounded in the Source of Truth; distractors are SoT-anchored, never invented. RI.5.4 and RI.5.7 are not used, no RL codes are used, and there are no "from the text" snippet boxes.
- **Genre & credits:** Eruption! = Narrative Nonfiction (by Elizabeth Rusch, photographs by Tom Uhlman; the true story of the real VDAP/USGS scientists and the real 1991 Mount Pinatubo eruption); Between the Glacier and the Sea = Media: Video (from the Valdez Museum; the real 1964 Great Alaska Earthquake, magnitude 9.2); Quaking Earth, Racing Waves = Informational Text (by Rachel Young; the real geologist Kerry Sieh and the real 2004 Indian Ocean tsunami); Hurricanes = Informational Text (by Alvin and Virginia Silverstein and Laura Silverstein Nunn; accurate hurricane meteorology). Only the OCR-excerpted text was used; no outside facts, figures, or dates were added.
- **IB (Worksheet C):** the PYP concepts — Causation, Perspective, Systems, Responsibility, Inquiry — follow the Source-of-Truth themes of this science module. The Essential Question text on Q5 matches the workbook word for word. Korean teacher guidance appears in Section C only; Sections A, B, and D are in English. No model answer adds a fact, statistic, or date outside the source — the IB model answers describe only what the OCR-excerpted selections actually show.
- **MAP (Worksheet D):** four NEW passages (200 / 203 / 205 / 206 words), RIT 205→225, Lexile 640L–940L. None recall the four anchor selections; each is a different informational text (a volcano warning, what makes the ground shake, a tsunami warning, how a hurricane forms), in keeping with this science-heavy module. All four are informational (RI) — there is no literary passage, because the module has no literary selection.
- **Answer balance:** 83 multiple-choice items distribute as 21 "A" / 21 "B" / 21 "C" / 20 "D"; every meaning is preserved from the Source of Truth.

MAP Mini-Passage Word Counts (Worksheet D)

- Passage 1 · The Mountain That Smoked · **200 words** · Lexile 640–720L · RIT 205–211
- Passage 2 · What Makes the Ground Shake? · **203 words** · Lexile 720–820L · RIT 211–217
- Passage 3 · The Wave That Saved a Town · **205 words** · Lexile 760–860L · RIT 217–222
- Passage 4 · How a Hurricane Is Born · **206 words** · Lexile 820–940L · RIT 219–225

Word counts match the workbook passage meta exactly. All four within target band 160–230 words; RIT 205→225; Lexile 640L–940L. Each passage is informational (science-focused), matching this module's science focus. Passages are NEW — no recall of the four anchor selections.

What This Module Does NOT Contain (Guardrails)

- No "Power Words" label — the G5 source says "Critical Vocabulary."
- No Worksheet E — the V3 workflow uses four worksheets (A · B · C · D).
- The module quote is by Dave Eggers and is the EPIGRAPH only — it is NOT a story author. Dave Eggers is a real, living American author.
- "Who Studies Natural Disasters?" is the Big-Idea Short Read / Letter (no Critical Vocabulary panel) and is excluded from the per-selection units; it carries the 4 Big Idea Words instead. Its author is unverified — no byline in the source (the in-text letters are signed by fictional correspondents Robin Thierssen and Professor Judy Melendez).
- Critical Vocabulary counts are $8 / 4 / 7 / 8 = 27$, not evened out. Big Idea (4) + Critical (27) = 31 total. Between the Glacier and the Sea genuinely has the shortest panel (4 words), expected for a short video selection.
- One of the four main selections is a Media: Video (Between the Glacier and the Sea). It is still a vocabulary-bearing main selection and is classified RI for CCSS. RI.5.7 (multimedia integration) would be its natural fit but is EXCLUDED per spec, so it is tested with other RI codes.
- The opener Get Curious video is "Nature's Dark Side" — a separate opener video, NOT the in-module video selection.
- The topic strand is "SCIENCE CONNECTION: Natural Disasters." Module 2 "What a Story" (Airborn, The Secret Garden, The Miracle of Spring, The Poem That Will Not End) and Module 4 are OUT OF SCOPE and are not conflated with any Module-3 selection.
- No outside statistics or dates were invented. The only figures used are those verified in the SoT (the 1991 Mount Pinatubo eruption with more than 20,000 lives saved; the 1964 Alaska earthquake's 9.2 magnitude; earthquakes near Tello Island about every 200 years; the four hurricane stages; the Saffir-Simpson Category 1–5 scale).
- Performance Task = "Write an Opinion Essay."

A Worksheet A — Vocabulary · Answer Key

31 items · CCSS L.5.4 / L.5.5 / L.5.6

BIG IDEA WORDS

Q1

L.5.6

notable

C

Meaning: if something is notable, it is worth noticing.

Source: SoT Big Idea Word — Vocabulary Card 3.1; inline myNotes. *Distractors* A/B/D give too small / a map / a secret.

Q2

L.5.6

tremor

D

Meaning: a tremor is a small earthquake or uncontrolled shaking in a body part. **Source:** SoT Big Idea Word 3.2; inline myNotes. *Distractors* give a wave / a wind / thunder.

Q3

L.5.5

TRAP

spontaneous

A

Meaning: a spontaneous action happens naturally and isn't planned. **Source:** SoT Big Idea Word 3.3; inline myNotes. **Trap:** B/C bait the opposite (planned ahead / scheduled).

Q4

L.5.6

hazard

C

Meaning: a hazard is a danger. **Source:** SoT Big Idea Word 3.4; inline myNotes. *Distractors* give a safe place / advice / a tool.

CRITICAL VOCABULARY · ERUPTION!

Q5

L.5.4

seismographs

D

Evidence: SoT — "Seismographs are instruments that measure and record details about earthquakes" (inline myNotes). Body: "the seismographs recorded the first earthquake." *Distractors* give trucks / maps / photos.

Q6

L.5.4

evacuation

B

Evidence: SoT — "An evacuation is the act of moving from a dangerous area to a safer one" (inline myNotes). Body: "moved to evacuation camps." *Distractors* give building / repairing / counting.

Q7

L.5.4

reservoir

C

Evidence: SoT — "A reservoir is a place where a supply of something is collected" (inline myNotes). Body: "the reservoir of melted rock and gas under the volcano." *Distractors* give fire / a path / a sign.

Q8

L.5.4

conferring

B

Evidence: SoT — "If you are conferring with someone, you are discussing an idea or trying to make a decision" (inline myNotes). Body: "After conferring with VDAP scientists." *Distractors* give arguing / ignoring / drawing.

Q9

L.5.5

TRAP

consequences

A

Evidence: SoT — "Consequences are the outcomes or effects of events" (inline myNotes). Body: "The consequences ... weighed heavily on their minds." **Trap:** B baits the opposite (the causes that come before).

Q10

L.5.4

widespread

A

Evidence: SoT — "If something is widespread, it happens over a large area or among many people" (inline myNotes). Body: "the widespread devastation ... wrought by this volcano." *Distractors* give one spot / underground / at night.

Q11

L.5.5

TRAP

alarming

C

Evidence: SoT — "Something that is alarming makes you worry that something bad may happen" (inline myNotes). Body: "making alarming blocks of solid ink." **Trap:** A/B bait the opposite (calm and safe / sleepy).

Q12

L.5.4

victim

D

Evidence: SoT — "If you fall victim to something, you suffer or die because of it" (inline myNotes). Body: "before they themselves fell victim to an eruption." *Distractors* give teach / escape / measure.

CRITICAL VOCABULARY · BETWEEN THE GLACIER AND THE SEA

Q13

L.5.5

TRAP

prior

D

Evidence: SoT — "Prior means coming before, in time or order" (Critical Vocabulary box, video selection). **Trap:** A baits the opposite (coming after everything else).

Q14

L.5.4

magnitude

B

Evidence: SoT — "Magnitude refers to the greatness of size or rank; in geology, magnitude refers to the rank of an earthquake." Possible response: "The 9.2 magnitude was the largest ever recorded in North American history." *Distractors give color / time / a name.*

Q15

L.5.5

TRAP

literally

A

Evidence: SoT — "If you say something literally happened, that means it actually happened, and you aren't exaggerating or using a metaphor." **Trap:** B/C bait the opposite (make-believe / an exaggeration).

Q16

L.5.4

maintenance

A

Evidence: SoT — "The maintenance of something is the act of caring for it and repairing it when needed." *Distractors give selling / hiding / measuring.*

CRITICAL VOCABULARY · QUAKING EARTH, RACING WAVES

Q17

L.5.4

parallel

B

Evidence: SoT — "If two or more things are parallel to each other, they move in the same direction and never touch or intersect" (inline myNotes). *Distractors give cross / spin / melt.*

Q18

L.5.4

lateral

C

Evidence: SoT — "If something moves in a lateral way, it moves side to side" (inline myNotes). *Distractors give up / down / in a circle.*

Q19

L.5.5

TRAP

destruction

C

Evidence: SoT — "Destruction is the act of destroying or ruining something" (inline myNotes). Body: "[the force] causes the destruction." **Trap:** A baits the opposite (building something new).

Q20

L.5.4

triggered

A

Evidence: SoT — "If you triggered an action or event, you did something to start it" (inline myNotes). Body: "the tsunami triggered by the earthquake." *Distractors give stop / watch / report.*

Q21

L.5.4

thrust

A

Evidence: SoT — "To thrust is to push something with great force" (inline myNotes). *Distractors give pull gently / hold still / measure.*

Q22

L.5.4

radiate

D

Evidence: SoT — "To radiate is to spread out in waves or rays" (inline myNotes). *Distractors give sink / freeze / stop.*

Q23

L.5.5

TRAP

modified

D

Evidence: SoT — "A modified version of something is a revised, or changed, version" (inline myNotes). Body: "the modified town is better prepared for tsunamis." **Trap:** A/B bait the opposite (an exact copy / the very first version).

Q24

L.5.4

surge

B

Evidence: SoT — "A surge is a sudden powerful movement forward or upward" (inline myNotes). Body: "part of Fitow's storm surge." *Distractors give a drip / a still pool / dust.*

Q25

L.5.5

TRAP

eventually

C

Evidence: SoT — "To state that something will happen eventually means that it will happen at some time, usually after a series of other events" (inline myNotes). Body: "eventually it dies out." **Trap:** A/B bait the opposite (this very second / never).

Q26

L.5.4

regions

B

Evidence: SoT — "Regions are areas" (inline myNotes). Body: "More hurricanes hit some regions than others." *Distractors give weather / seasons / boats.*

Q27

L.5.4

mobile

A

Evidence: SoT — "Something that is mobile is able to move or be moved easily" (inline myNotes). Body: "Many of the buildings there were mobile homes." *Distractors give underground / glass / a color.*

Q28

L.5.5

TRAP

anchored

B

Evidence: SoT — "Something that is anchored is firmly attached to something else or weighed down so it won't move easily" (inline myNotes). Body: "not well anchored to the ground." **Trap:** A/C bait the opposite (loose / floating).

Q29

L.5.4

foundations

D

Evidence: SoT — "Foundations are the base pieces houses and buildings are built upon" (inline myNotes). Body: "remove the soil around the foundations of houses." *Distractors give roofs / windows / fences.*

Q30

L.5.4

category

B

Evidence: SoT — "A category is a group of things that are similar to each other in some way" (inline myNotes). Body: "rank ... from Category 1 ... to Category 5." *Distractors give a flash / a tool / a path.*

Q31

L.5.4

track

C

Evidence: SoT — "To track something is to watch it and see where it moves and how it changes" (inline myNotes). Body: "scientists can track hurricanes better." *Distractors give push / hide / repair.*



Worksheet B — Reading Comprehension · Answer Key

32 items · CCSS RI.5.2 / .3 / .5 / .6 / .8

Selection 1 · Eruption! Volcanoes and the Science of Saving Lives

NARRATIVE NONFICTION · 8 ITEMS

Q1

RI.5.2

C

Central Idea/Gist: a team of volcano scientists monitors Mount Pinatubo and persuades people to evacuate, saving many lives (SoT key text + theme). *Others name a different selection.*

Q2

RI.5.3

A

Relationships: SoT — after an earlier eruption "horrified" Andy Lockhart, he joined the Volcano Disaster Assistance Program (VDAP). *Others off-text.*

Q3

RI.5.8

D

Reasons/Evidence: SoT — "The consequences—a costly false evacuation or tragic loss of life—weighed heavily on their minds." *Others off-text.*

Q4

RI.5.3

B

Relationships: SoT — "the seismographs recorded the first earthquake"; the instruments revealed building pressure inside the volcano. *Others off-text.*

Q5

RI.5.5

D

Text Structure: SoT GENRE STUDY — narrative nonfiction presents events in chronological order, telling a true story as it happened. *Others name the wrong structure.*

Q6

RI.5.6

C

Purpose: SoT — primary-source quotations and photographs help readers understand the real events and feel the danger. *Others off-purpose.*

Q7

RI.5.3

A

Relationships: SoT — after the scientists conferred and warned the military, people were moved away and "more than 20,000 lives were saved." *Others off-text.*

Q8

RI.5.2

B

Central Idea: by monitoring a volcano and making hard decisions, scientists can predict an eruption and save thousands of lives (SoT theme). *The other choices are false or unrelated.*

Selection 2 · Between the Glacier and the Sea: The Alaska Earthquake

MEDIA: VIDEO · 8 ITEMS

Q9

RI.5.2

B

Central Idea/Gist: the 1964 Great Alaska Earthquake and its effect on the town of Valdez (SoT genre + characters note). *Others name a different selection.*

Q10

RI.5.6

D

Purpose: SoT GENRE STUDY — "First-person accounts are sometimes featured to help viewers better understand an event or experience." *Others off-purpose.*

Q11

RI.5.5

A

Media Structure: SoT — a narrator explains what is happening while animations show plate movement, so viewers both hear and see the ideas. *Others off-text.*

Q12

RI.5.2

C

Key Detail/Idea: SoT Possible response — "five minutes of violent shaking ... The 9.2 magnitude was the largest ever recorded in North American history." *Others off-text.*

Q13

RI.5.3

C

Relationships: SoT — much of Valdez was destroyed, and the earthquake led to dangerous oil fires. *Others off-text.*

Q14

RI.5.3

A

Word/Idea: SoT — "magnitude refers to the rank of an earthquake," its greatness of size; it shows how strong the quake was. *Others off-text.*

Q15

RI.5.6

B

Purpose: SoT GENRE STUDY — an informational video presents information to inform viewers about the event's causes and human impact. *Others off-purpose.*

Q16

RI.5.2

D

Central Idea: firsthand accounts and visuals together help people understand the power and damage of a major earthquake (SoT theme). *The other choices are false or unrelated.*

Selection 3 · Quaking Earth, Racing Waves

INFORMATIONAL TEXT · 8 ITEMS

Q17

RI.5.2

D

Central Idea/Gist: how earthquakes and tsunamis form, and how a scientist used clues to predict one (SoT key text). *Others name a different selection.*

Q18

RI.5.5

B

Text Structure: SoT GENRE STUDY — organized under subheadings, with diagrams and a map that tell more about the topic. *Others name the wrong structure.*

Q19

RI.5.3

C

Relationships: SoT — Kerry Sieh "looked at the growth patterns of Porites coral heads near the fault" to pinpoint the dates of past earthquakes. *Others off-text.*

Q20

RI.5.3

A

Relationships: SoT — "a tsunami is started by a disturbance, such as an undersea earthquake, that shifts water at the ocean floor." *Others off-text (most waves come from wind).*

Q21

RI.5.8

A

Reasons/Evidence: SoT — by reading the corals, Sieh found earthquakes occurred about every 200 years near the island. *Others contradict the text.*

Q22

RI.5.3

C

Relationships: SoT — "the modified town is better prepared for tsunamis"; the changes made it safer. *Others off-text.*

Q23

RI.5.6

D

Purpose/Features: SoT — diagrams (coral growth, plate boundaries) and a map help readers picture how the earth moves and where events take place. *Others off-purpose.*

Q24

RI.5.2

B

Central Idea: by studying evidence such as corals and plate movement, scientists can predict earthquakes and tsunamis and help people prepare (SoT theme). *The other choices are false or unrelated.*

Selection 4 · Hurricanes: The Science Behind Killer Storms

INFORMATIONAL TEXT · 8 ITEMS

Q25

RI.5.2

B

Central Idea/Gist: how hurricanes form, what dangers they bring, and how people can stay safe (SoT characters/theme note). *Others name a different selection.*

Q26

RI.5.5

D

Text Structure: SoT GENRE STUDY — headings, charts, a map, and sidebar features help readers find the central ideas. *Others name the wrong structure.*

Q27

RI.5.3

C

Relationships: SoT — "A hurricane goes through a series of four stages ...: tropical disturbance, tropical depression, tropical storm, and hurricane." *Others off-text.*

Q28

RI.5.8

A

Reasons/Evidence: SoT — mobile homes "are not well anchored to the ground," so strong winds can move or damage them easily. *Others off-text.*

Q29

RI.5.3

C

Relationships: SoT — "people are more likely to die from a storm surge"; the surge pushes a huge, sudden rush of seawater onto the land. *Others off-text.*

Q30

RI.5.3

A

Relationships: SoT — the "Saffir-Simpson Hurricane Scale" ranks storms into categories so people know how strong and dangerous a storm is. *Others off-text.*

Q31

RI.5.6

D

Purpose: SoT — the authors explain the science of hurricanes so readers understand the danger and can prepare and stay safe. *Others off-purpose.*

Q32

RI.5.2

TRAP

B

Central Idea: understanding how hurricanes form and what damage they cause helps people prepare and stay safe (SoT theme).
Trap: the other choices are true details, not the central idea.



Worksheet C — IB Critical Thinking · Answer Key

5 questions · PYP Key Concepts · DOK 2–4

Q1

Causation

DOK 3

In "**Eruption!**," scientists watched for warning signs before Mount Pinatubo erupted. Show the cause and the effect, then explain how their work helped people.

MODEL ANSWER

What the Scientists Noticed: The volcano team, including Andy Lockhart, used seismographs that recorded earthquakes inside Mount Pinatubo. The instruments showed that pressure was building as magma moved beneath the volcano — clear warning signs that it might erupt. **What They Decided to Do:** After conferring with one another, the scientists warned the military and urged an evacuation, so that people would be moved away from the danger before the volcano blew. **Why the decision was hard:** The consequences of being wrong were serious either way. A false alarm would be costly, but waiting too long could mean a tragic loss of life, so the choice weighed heavily on the scientists' minds and hearts. **What this shows:** Because they read the warning signs and made a brave decision, more than 20,000 lives were saved when Mount Pinatubo finally erupted. Their scientific knowledge turned a deadly event into one that many people survived.

KEYWORDS (3)

cause = seismographs showed earthquakes and rising pressure inside the volcano

effect = scientists warned the military and people evacuated

result = more than 20,000 lives were saved

SCORING NOTES

- Full credit (3): the warning signs (cause) + the decision to evacuate (effect) + how it helped, using the word *consequences*.
- Partial (2): names a cause and an effect but no clear explanation of the hard decision.
- Minimal (1): only "the volcano erupted" with no science.

TEACHING STEPS

Step 1 · Two boxes.

강사: "Label two boxes: *What the Scientists Noticed*, *What They Decided to Do*."

코멘트. 상자 두 칸 — 과학자가 관찰한 것·결정한 것. 각 칸 채우기.

Step 2 · Find the cause.

강사: "Underline the sentence about the seismographs and the earthquakes."

코멘트. 지진계와 지진을 설명하는 문장에 밑줄.

Step 3 · Cause-effect sentence.

강사: "Finish: 'Because the scientists saw ____, they ____.'"

코멘트. 문장 완성 — '과학자들은 ____을 보았기 때문에 ____했다.'

STUDENT PROMPTS

- "What warning signs did the seismographs show?"

지진계는 어떤 경고 신호를 보여 줬을까?

- "Why was it risky to call for an evacuation?"
대피령을 내리는 것이 왜 위험한 결정이었을까?
- "How did the scientists' choice change what happened?"
과학자들의 선택이 결과를 어떻게 바꿨을까?

DIFFERENTIATION

STRUGGLING Match 2 cards ("earthquakes inside the volcano" / "people moved to safety") to cause and effect.

카드 2장 — '화산 속 지진'과 '사람들 대피'를 원인·결과와 매칭.

ADVANCED Explain why a scientist must sometimes act before being 100% sure.

과학자가 100% 확신 전에 행동해야 하는 이유 설명.

COMMON MISCONCEPTIONS

PITFALL "The scientists stopped the eruption." They could not stop it; they predicted it and moved people to safety.

화산 폭발을 막은 게 아님 — 예측해 사람들을 대피시킴.

PITFALL "This is a made-up story." It is a true account of the real 1991 Mount Pinatubo eruption and real scientists.

지어낸 이야기가 아님 — 1991년 피나투보 화산의 실화.

LEARNER PROFILE LINK

KNOWLEDGEABLE The scientists used what they knew about volcanoes to read the warning signs.

화산 지식으로 경고 신호를 읽음 — Knowledgeable.

PRINCIPLED They made a hard, responsible choice to protect people's lives.

사람들을 지키기 위한 책임 있는 선택 — Principled.

KOREAN BRIDGE

표현 "The warning signs caused the scientists to act." → "경고 신호가 과학자들을 행동하게 만들었다."

인과(Causation) 핵심 문장.

단어 "seismographs / 지진계 — evacuation / 대피 — conferring / 의논하는 — consequences / 결과"

핵심 단어 영-한.

In **"Between the Glacier and the Sea,"** show two tools the video uses to tell about the 1964 Alaska earthquake, then explain why firsthand accounts matter.

MODEL ANSWER

A Tool the Video Uses: The video uses a narrator who explains what is happening on the screen, along with animations that show how the earth's plates moved during the earthquake. These visuals help viewers picture an event they cannot see in real life. **Another Tool the Video Uses:** The video also includes first-person accounts from people who lived through the earthquake. Hearing survivors describe the violent shaking and the damage in Valdez makes the event feel real. **Why firsthand accounts matter:** Facts alone can tell you that the earthquake had a 9.2 magnitude, but a survivor's own words help you understand what it was truly like to be there. The people who experienced it share feelings and details that a list of facts cannot. **What this shows:** By combining narration, animations, and firsthand accounts, the video helps viewers both understand the science and feel the human side of a major earthquake.

KEYWORDS (3)

tool 1 = narration with animations that show plate movement

tool 2 = first-person accounts from survivors

why it matters = firsthand voices help us understand what the event was really like

SCORING NOTES

- Full credit (3): two media tools + an explanation of why firsthand accounts help viewers understand.
- Partial (2): names two tools but no clear explanation.
- Minimal (1): only "it is a video" with no tool named.

TEACHING STEPS

Step 1 · Two columns.

강사: "Make two columns: Tool 1 and Tool 2."

코멘트. 두 칸 — 도구 1, 도구 2. 각각 영상 기법 적기.

Step 2 · Spot the voices.

강사: "Mark where a survivor speaks and where the narrator explains."

코멘트. 생존자가 말하는 부분과 해설자가 설명하는 부분 표시.

Step 3 · Perspective sentence.

강사: "Finish: 'A survivor's words help me understand ____.'"

코멘트. 문장 완성 — '생존자의 말은 내가 ____을 이해하도록 돕는다.'

STUDENT PROMPTS

- "What do the animations help you see?"
애니메이션은 무엇을 보게 해 줄까?
- "How is hearing a survivor different from reading a fact?"
생존자 이야기를 듣는 것은 사실을 읽는 것과 어떻게 다를까?
- "Why might a video use both pictures and voices?"

영상은 왜 그림과 목소리를 함께 쓸까?

DIFFERENTIATION

STRUGGLING Match 2 cards ("a narrator explains" / "a survivor remembers") to the two tools.

카드 2장 — '해설자 설명'과 '생존자 회상'을 두 도구와 매칭.

ADVANCED Explain how a video can teach in ways a plain article cannot.

영상이 일반 글이 못 하는 방식으로 가르치는 점 설명.

COMMON MISCONCEPTIONS

PITFALL "The survivors are actors." They are real people sharing real memories of the 1964 earthquake.

생존자는 배우가 아님 — 1964년 지진의 실제 경험자.

PITFALL "Animations are just for fun." They explain how the plates moved, helping viewers understand the cause.

애니메이션은 재미용이 아님 — 지각판 움직임을 설명.

LEARNER PROFILE LINK

COMMUNICATORS Survivors share their experience so others can understand it.

경험을 나누는 생존자 — Communicator.

OPEN-MINDED Listening to many voices builds a fuller picture of the event.

여러 목소리를 들어 사건을 폭넓게 이해 — Open-minded.

KOREAN BRIDGE

표현 "Firsthand accounts help us understand an event." → "직접 겪은 사람의 이야기는 사건을 이해하도록 돕는다."

관점(Perspective) 핵심 문장.

단어 "prior / 이전의 — magnitude / 규모 — literally / 말 그대로 — maintenance / 정비"

핵심 단어 영-한.

In "**Quaking Earth, Racing Waves**," show the steps by which an undersea earthquake can lead to a tsunami, then explain how a scientist predicted the danger.

MODEL ANSWER

The Cause: Under the ocean, the earth's plates push against each other along a fault. When they suddenly move, an undersea earthquake shifts the water at the ocean floor. **What the Water Does:** That disturbance pushes a huge amount of water, which begins to radiate outward as waves. Unlike ordinary waves made by wind, these waves carry enormous energy across the ocean. **The Result:** As the waves reach shallow water near the shore, they rise into a towering tsunami that can cause great destruction when it strikes the land. **How a scientist predicted it:** The geologist Kerry Sieh studied the growth patterns of coral heads near the fault. The corals showed that earthquakes occurred about every 200 years, so he warned the villagers that a major earthquake and tsunami were due. **What this shows:** Because Sieh did not ignore the evidence in the corals, he understood the earth's system and could warn people before disaster struck.

KEYWORDS (3)

cause = an undersea earthquake shifts water at the ocean floor

water = the disturbance radiates outward and rises into a tsunami near shore

prediction = Sieh read coral growth to learn earthquakes came about every 200 years

SCORING NOTES

- Full credit (3): the cause + how the wave forms + the result + how Sieh predicted it, using the word *triggered*.
- Partial (2): names some steps but not the prediction, or no order.
- Minimal (1): only "an earthquake makes a wave" with no detail.

TEACHING STEPS

Step 1 · Three boxes.

강사: "Fill three boxes: *The Cause, What the Water Does, The Result.*"

코멘트. 세 칸 — 원인·물의 움직임·결과. 각 칸 채우기.

Step 2 · Find the clue.

강사: "Underline the sentence about Kerry Sieh reading the corals."

코멘트. 케리 시가 산호를 읽는 문장에 밑줄.

Step 3 · System sentence.

강사: "Finish: 'An undersea earthquake triggered ____, which led to ____.'"

코멘트. 문장 완성 — '해저 지진은 ____을 일으켰고, 그것은 ____로 이어졌다.'

STUDENT PROMPTS

- "What starts a tsunami, according to the text?"
본문에 따르면 쓰나미는 무엇으로 시작될까?
- "How is a tsunami different from an ordinary wave?"
쓰나미는 보통 파도와 어떻게 다를까?

- "How did the corals help Kerry Sieh warn people?"

산호는 케리 시가 사람들에게 경고하도록 어떻게 도왔을까?

DIFFERENTIATION

STRUGGLING Put 3 cards in order: "earthquake," "water pushed," "wave hits shore."

카드 3장 순서대로 — '지진'·'물이 밀림'·'파도가 해안에 도달'.

ADVANCED Explain how studying a pattern (every 200 years) lets a scientist predict the future.

200년 주기 같은 패턴 연구가 어떻게 미래 예측을 가능하게 하는지 설명.

COMMON MISCONCEPTIONS

PITFALL "Tsunamis are caused by wind." Most waves are, but a tsunami is started by an undersea disturbance such as an earthquake.

쓰나미는 바람 때문이 아님 — 해저 지진 같은 충격으로 시작.

PITFALL "Earthquakes cannot be predicted at all." Sieh used coral evidence to warn that one was likely.

지진을 전혀 예측 못 하는 건 아님 — 시는 산호 증거로 경고함.

LEARNER PROFILE LINK

INQUIRERS Sieh kept asking what the corals could reveal about the past.

산호가 과거에 대해 무엇을 알려 주는지 캐물음 — Inquirer.

THINKERS Connecting coral patterns to earthquakes is real scientific thinking.

산호 패턴과 지진을 연결하는 과학적 사고 — Thinker.

KOREAN BRIDGE

표현 "An undersea earthquake can trigger a tsunami." → "해저 지진은 쓰나미를 일으킬 수 있다."

시스템(Systems) 핵심 문장.

단어 "parallel / 평행한 — destruction / 파괴 — triggered / 촉발된 — radiate / 퍼져 나가다 — modified / 수정된"

핵심 단어 영-한.

In "**Hurricanes**," show one danger a hurricane brings and one way people can stay safe, then explain why this knowledge matters.

MODEL ANSWER

A Danger a Hurricane Brings: A hurricane can bring a storm surge — a sudden, powerful rush of seawater pushed onto the land. The text explains that people are more likely to die from a storm surge than from the winds. Hurricanes also bring whipping winds and heavy rains. **A Way People Can Stay Safe:** People can prepare before a storm. For example, they can leave mobile homes, which are not well anchored to the ground, and move to a safer, sturdier place away from the surge. Families can make an emergency plan ahead of time. **Why this knowledge matters:** When people understand how a storm works and how strong it is — for instance, by checking a hurricane's category on the Saffir-Simpson scale — they can decide when to leave and how to protect themselves. **What this shows:** Understanding the science of hurricanes helps people take responsibility for their own safety and prepare before the storm arrives.

KEYWORDS (3)

danger = a storm surge pushes seawater onto the land

staying safe = leave unanchored mobile homes and make a plan before the storm

why it matters = knowing a storm's category helps people decide how to prepare

SCORING NOTES

- Full credit (3): one danger + one way to prepare + why the knowledge matters, using the word *category*.
- Partial (2): names a danger and a safety step but no clear explanation.
- Minimal (1): only "hurricanes are dangerous" with no detail.

TEACHING STEPS

Step 1 · Two columns.

강사: "Make two columns: A Danger / A Way to Stay Safe."

코멘트. 두 칸 — 위험 / 안전 대비 방법.

Step 2 · Find the danger.

강사: "Underline the sentence about the storm surge."

코멘트. 폭풍 해일(storm surge)에 관한 문장에 밑줄.

Step 3 · Safety sentence.

강사: "Finish: 'Knowing a storm's category helps people ____.'"

코멘트. 문장 완성 — '폭풍의 등급(category)을 아는 것은 사람들이 ____하도록 돕는다.'

STUDENT PROMPTS

- "Why is a storm surge so dangerous?"
폭풍 해일은 왜 그렇게 위험할까?
- "Why are mobile homes risky in a hurricane?"
이동식 주택은 허리케인에서 왜 위험할까?

- "How does knowing a storm's category help your family?"

폭풍의 등급을 아는 것이 가족에게 어떻게 도움이 될까?

DIFFERENTIATION

STRUGGLING Match 2 cards ("a storm surge" / "move to a safer place") to danger and safety.

카드 2장 — '폭풍 해일'과 '안전한 곳으로 이동'을 위험·안전과 매칭.

ADVANCED Explain how a warning scale (Category 1–5) helps people make good choices.

경고 등급(1~5)이 좋은 선택을 어떻게 돕는지 설명.

COMMON MISCONCEPTIONS

PITFALL "The wind is always the deadliest part." The text says people are more likely to die from a storm surge.

바람이 늘 가장 치명적인 건 아님 — 본문은 폭풍 해일을 더 위험하다고 설명.

PITFALL "Nothing can be done about a hurricane." Preparing and following the category warnings helps keep people safe.

대비할 수 없는 게 아님 — 준비와 등급 경고가 안전을 도움.

LEARNER PROFILE LINK

PRINCIPLED Preparing for a storm is taking responsibility for safety.

폭풍 대비는 안전에 대한 책임 — Principled.

KNOWLEDGEABLE Knowing how hurricanes work helps people make safe choices.

허리케인 원리를 알면 안전한 선택 가능 — Knowledgeable.

KOREAN BRIDGE

표현 "Understanding the science helps people prepare." → "과학을 이해하면 사람들이 대비할 수 있다."

책임(Responsibility) 핵심 문장.

단어 "surge / 해일 — regions / 지역 — mobile / 이동식의 — anchored / 고정된 — category / 등급 — track / 추적하다"

핵심 단어 영-한.

Essential Question: How can learning about natural disasters make us safer? Use ideas from TWO of the selections in this module to answer.

MODEL ANSWER

Learning about natural disasters helps people predict them and prepare, so fewer lives are lost. In **"Eruption!"**, scientists used seismographs to watch Mount Pinatubo. When the instruments showed building pressure, the team conferred and warned the military, and people were evacuated before the volcano blew — more than 20,000 lives were saved. In **"Quaking Earth, Racing Waves,"** the geologist Kerry Sieh studied the growth of coral heads near a fault. The corals showed that earthquakes happened about every 200 years, so he could warn villagers that a tsunami-producing earthquake was likely. **My Big Idea:** As Dave Eggers said, "Yes, a dark time passed over this land, but now there is something like light." When people learn how disasters work — by reading seismographs or coral records — they can predict the danger and act in time. So learning about natural disasters makes us safer because knowledge turns a deadly surprise into a danger we can prepare for.

KEYWORDS (3)

Eruption! = seismographs warned of pressure, so people evacuated and lives were saved

Quaking Earth, Racing Waves = coral records helped Sieh predict an earthquake and tsunami

Big Idea: learning how disasters work lets people predict the danger and prepare in time

SCORING NOTES

- Full credit (4): two selections named + how learning about each disaster helps keep people safe + a Big Idea answering the EQ.
- Partial (3): two selections + safety ideas, Big Idea weak.
- Developing (2): one selection only.
- Minimal (1): generic ("disasters are dangerous") without naming a selection.

SYNTHESIS FRAME

SELECTION 1 IDEA

Name the disaster and how learning about it keeps people safe:

Eruption! = seismographs warn of an eruption so people evacuate; Between the Glacier and the Sea = survivor accounts and visuals teach how an earthquake strikes; Quaking Earth, Racing Waves = coral and plate evidence predict earthquakes and tsunamis; Hurricanes = the category scale warns how strong a storm is.

CONNECTING WORD

BOTH / WHILE — joins two selections (for example Eruption! WHILE Quaking Earth, Racing Waves) into one Big Idea answering "How can learning about natural disasters make us safer?"

SELECTION 2 IDEA

A different disaster than Selection 1 — so the two show two different ways that scientific knowledge helps protect people.

REAL-WORLD CONNECTION

KOREA 자연재해를 공부하면 예측하고 대비할 수 있어 피해를 줄인다 — 화산·지진·쓰나미·허리케인 모두 과학 지식(지진계·산호 기록·등급 척도)으로 미리 경고할 수 있다. 안다는 것이 곧 안전이다.

우리가 아는 재해를 EQ와 연결하기.

PERSONAL What warning signs or safety plans does YOUR family know for an emergency?

너의 가족은 비상 상황을 위해 어떤 경고 신호나 안전 계획을 알고 있니?

RUBRIC

LEVEL	WHAT IT LOOKS LIKE
Exceeding (4)	Names 2 selections + how learning about each disaster keeps people safe + a Big Idea answering the EQ.
Meeting (3)	Names 2 selections + a safety idea for each + a Big Idea.
Approaching (2)	Names 2 selections but only retells the events — no clear answer to the EQ.
Beginning (1)	Generic ("disasters are scary") without text evidence.

Watch out: Many G5 students will just retell one disaster (a volcano erupted; a big wave came) and stop. Push them to explain HOW learning about the disaster — seismographs, coral records, survivor accounts, the category scale — helps people predict it and stay safe, and to write a Big Idea. The EQ asks how knowledge makes us SAFER, not just what happened.

TEACHING STEPS

Step 1 · Two selection cards.

강사: "Pick 2 selections. For each, name the disaster it explains."

코멘트. 선택지 둘 — 각각 설명하는 재해를 적기.

Step 2 · Name the safety idea.

강사: "Under each, write one way learning about it keeps people safe."

코멘트. 각각 아래 — 그것을 배우면 어떻게 안전해지는지 하나 적기.

Step 3 · Big Idea sentence.

강사: "Write one sentence answering the EQ. Use both selections."

코멘트. EQ에 답하는 한 문장 — 두 선택지 모두 사용.

STUDENT PROMPTS

- "What disaster does each selection you chose explain?"
고른 선택지마다 어떤 재해를 설명하니?
- "How did scientists use evidence to warn people?"
과학자들은 증거를 이용해 사람들에게 어떻게 경고했을까?
- "Why does Dave Eggers's quote fit this module?"
데이브 에거스의 인용문이 왜 이 모듈에 어울릴까?

DIFFERENTIATION

STRUGGLING Sentence frame: "___ explains ___. Learning about it keeps people safe by ___. ___ explains ___. It keeps people safe by ___. Knowledge makes us safer."

문장 골격 제공.

ADVANCED Use THREE selections and sort them by disaster (volcano / earthquake / tsunami / hurricane).

선택지 셋 사용 — 재해별로 분류.

COMMON MISCONCEPTIONS

PITFALL "Retelling what happened answers the EQ." The EQ asks HOW learning makes us safer, not just what the disaster was.

사건 줄거리만으로는 EQ 답이 아님 — '배움이 어떻게 안전하게 하는지'를 묻음.

PITFALL "The quote is by a story author." It is the module EPIGRAPH, by Dave Eggers, not a story author.

인용문은 이야기 작가의 말이 아님 — 모듈 에피그래프(데이브 에거스).

LEARNER PROFILE LINK

INQUIRERS Asking how knowledge keeps us safe is the heart of inquiry.

지식이 어떻게 안전을 지키는지 묻는 것이 탐구의 핵심 — Inquirer.

THINKERS Building one Big Idea from two disasters is real critical thinking.

두 재해로 하나의 Big Idea를 세움 — Thinker.

KOREAN BRIDGE

표현 "Learning about natural disasters helps people predict them and stay safe." → "자연재해를 배우면 예측하고 안전을 지킬 수 있다."

M3 마무리 문장 — 학생 글쓰기 핵심.

인용 "Yes, a dark time passed over this land, but now there is something like light." — Dave Eggers → "그렇다, 어두운 시간이 이 땅을 지나갔지만, 이제 빛과 같은 것이 있다."

에피그래프 — 재난 이후의 희망.

MAP Mini-Passage Word Counts (Worksheet D)

- Passage 1 · The Mountain That Smoked · **200 words** · Lexile 640–720L · RIT 205–211
- Passage 2 · What Makes the Ground Shake? · **203 words** · Lexile 720–820L · RIT 211–217
- Passage 3 · The Wave That Saved a Town · **205 words** · Lexile 760–860L · RIT 217–222
- Passage 4 · How a Hurricane Is Born · **206 words** · Lexile 820–940L · RIT 219–225

All four passages within target band 160–230 words; RIT 205→225; Lexile 640L–940L. Word counts match the workbook passage meta exactly. Each passage is informational (science-focused), matching this module's science focus. Passages are NEW — no recall of the four anchor selections.

PASSAGE 1 · THE MOUNTAIN THAT SMOKED

Q1 RIT 205

C

Lexile 640L · Key Detail. "It was a volcano, and deep inside it, melted rock was slowly building up." *Distractors A/B/D off-text.*

Q2 RIT 207

B

Lexile 660L · Cause & Effect. "The signs told her that pressure was building ... the volcano could erupt soon." *Trap A/C/D off-text.*

Q3 RIT 208

A

Lexile 680L · Key Detail. "She set up instruments that measured the tiny earthquakes, and she watched the steam." *Trap B/C/D off-text.*

Q4 RIT 210

D

Lexile 700L · Vocabulary. "the ground ... began to shake with small tremors" — tremors = small shaking movements of the ground. *Trap A/B/C take the wrong meaning.*

Q5 RIT 211

B

Lexile 720L · Main Idea. The warning signs let the village evacuate in time — reading a volcano's warning signs can help people stay safe. *Trap A/C/D narrow or off-topic.*

PASSAGE 2 · WHAT MAKES THE GROUND SHAKE?

Q6 RIT 211

D

Lexile 720L · Key Detail. "it is broken into giant slabs of rock called plates ... like the pieces of a cracked eggshell." *Trap A/B/C off-text.*

Q7 RIT 213

A

Lexile 750L · Cause & Effect. "All that stored energy is released at once, and it radiates outward in waves" that shake the ground. *Trap B/C/D off-text.*

Q8 RIT 214

D

Lexile 780L · Text Structure. The passage explains the chain of causes that leads to the effect of an earthquake. *Trap A/B/C name the wrong structure.*

Q9 RIT 215

C

Lexile 800L · Vocabulary. "the size, or magnitude, of each earthquake" — magnitude = the size, or strength, of an earthquake. *Trap A/B/D take the wrong meaning.*

Q10 RIT 217

A

Lexile 820L · Main Idea. Moving plates cause earthquakes, and studying them helps people build safer buildings and prepare. *Trap B/C/D narrow or off-topic.*

PASSAGE 3 · THE WAVE THAT SAVED A TOWN

Q11 RIT 217

C

Lexile 760L · Key Detail. "if the sea ever pulls far away from the shore, run for high ground." *Trap A/B/D off-text.*

Q12 RIT 218

B

Lexile 790L · Cause & Effect. "The ocean was sliding backward ... She remembered her grandmother's words" and warned everyone. *Trap A/C/D off-text.*

Q13 RIT 219

B

Lexile 820L · Inference. "Some people stared, unsure" — they did not understand what the strange sight meant. *Trap A/C/D misread the moment.*

Q14 RIT 220

A

Lexile 840L · Cause & Effect. "a towering wall of seawater rushed in and swallowed the beach where they had all been standing." *Trap B/C/D off-text.*

Q15 RIT 222

A

Lexile 860L · Theme. A child's memory of an old warning saved the town — knowing the warning signs of danger can save lives. *Trap B/C/D trivialize it.*

PASSAGE 4 · HOW A HURRICANE IS BORN

Q16 RIT 219

D

Lexile 820L · Key Detail. "It starts over warm ocean water near the equator." *Trap A/B/C off-text.*

Q17 RIT 220

C

Lexile 850L · Cause & Effect. "once it reaches land ... it loses its supply of warm, moist air, and it begins to weaken." *Trap A/B/D off-text.*

Q18 RIT 221

D

Lexile 880L · Key Detail. "At the very center is a calm, clear area called the eye." *Trap A/B/C name the wrong part.*

Q19 RIT 223

B

Lexile 910L · Vocabulary. "As the warm water evaporates, it rises into the air" — evaporates = changes into a vapor and rises into the air. *Trap A/C/D take the wrong meaning.*

Q20 RIT 225

C

Lexile 940L · Main Idea. A hurricane forms over warm water and grows stronger there, and scientists track it to warn people in time. *Trap A/B/D narrow or off-topic.*